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(54) INFORMATION PROCESSING APPARATUS, DISPLAY CONTROL METHOD, AND RECORDING MEDIUM

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(57) ABSTRACT

An information processing apparatus includes a processor, in which, in a case of setting a first display region and a second display region on a display screen on which a display displays designated information, the first display region displaying a first portion of the information in a scrollable state and the second display region displaying a second portion of the information different from the first portion in a non-scrollable state, the processor changes an amount of displayable information in the first display region and an amount of displayable information in the second display region in accordance with a user operation of scrolling display in the first display region.

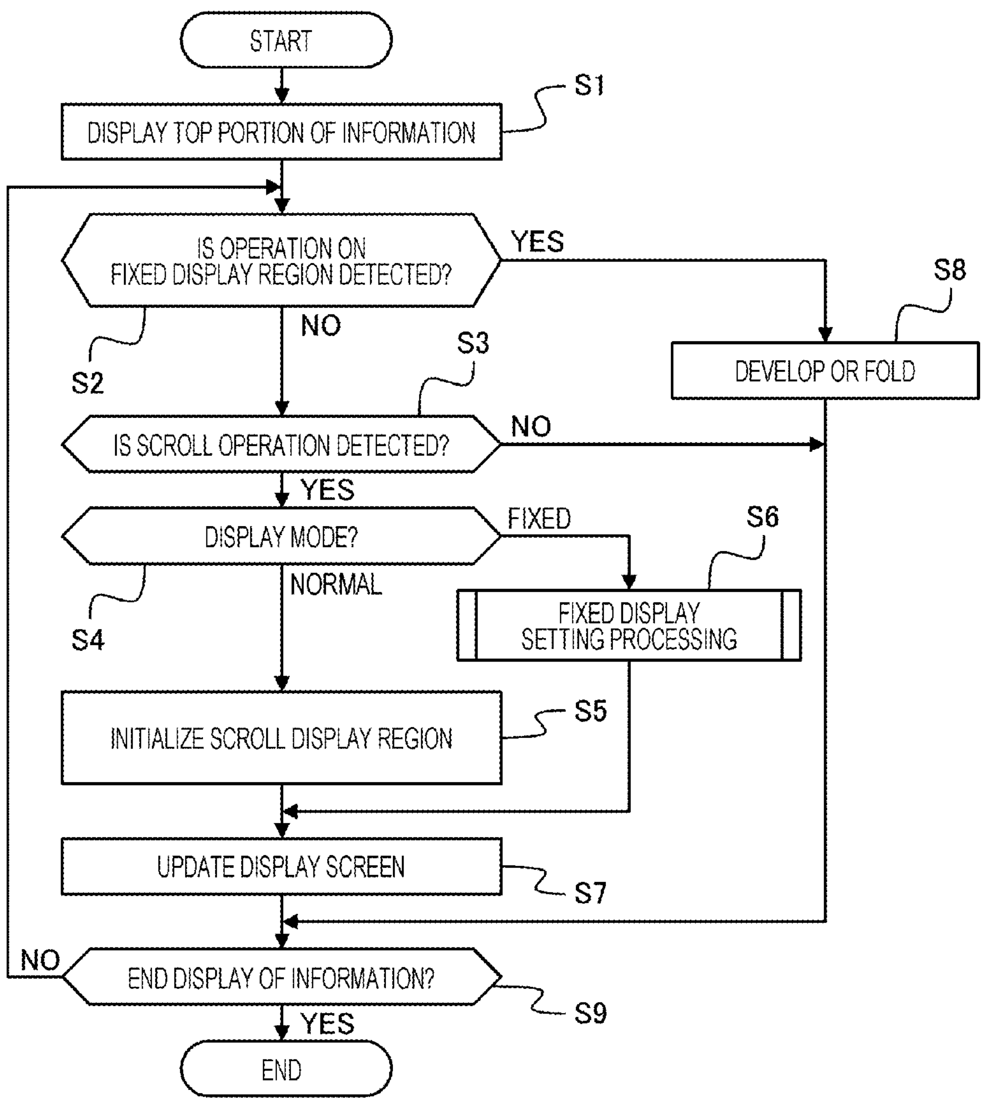


FIG. 1

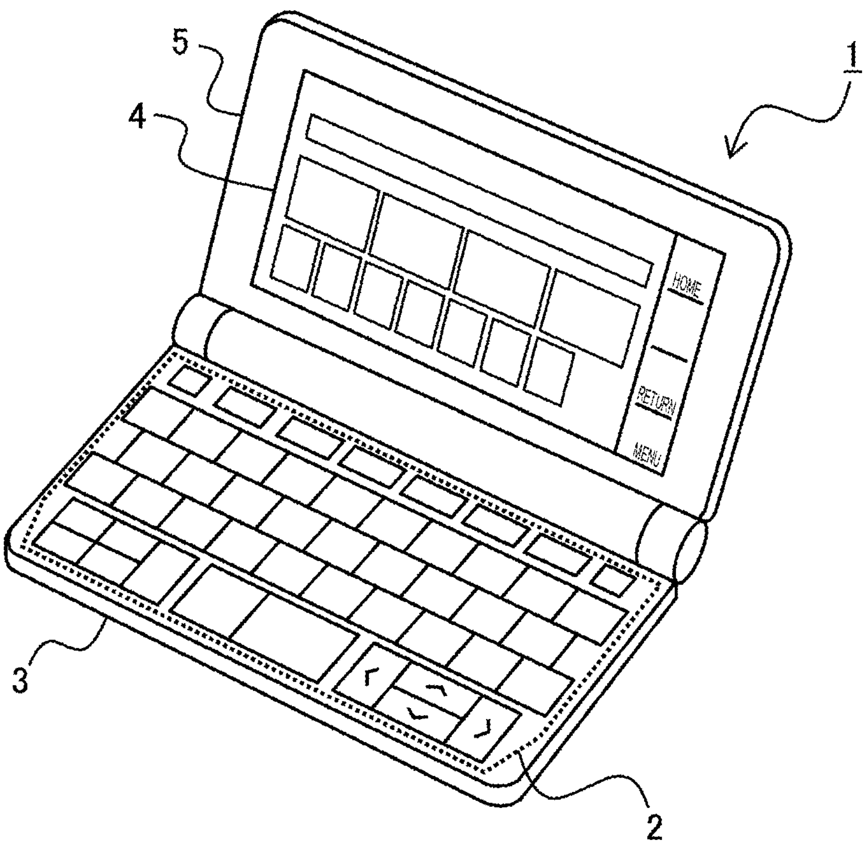


FIG. 2

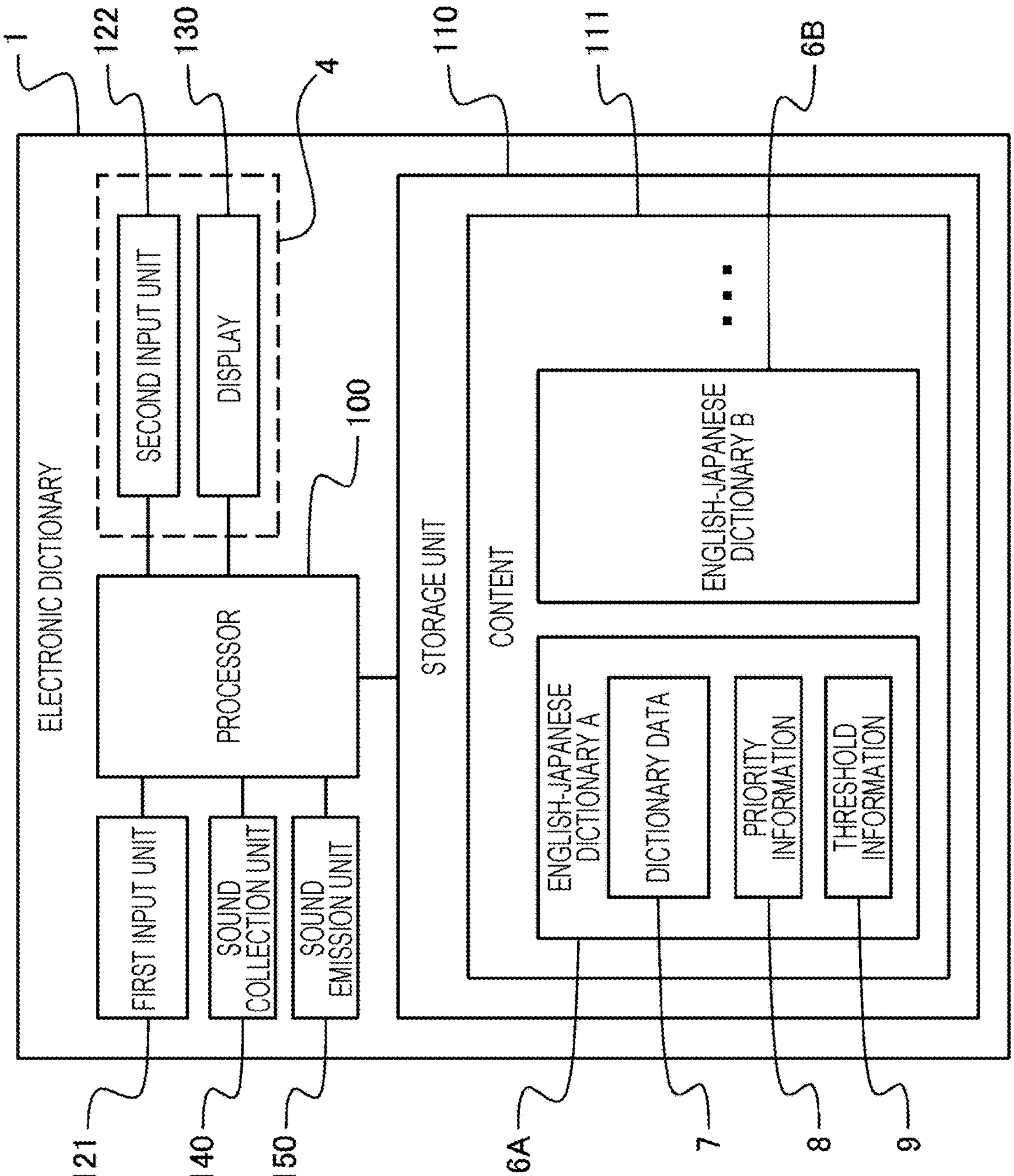

1
ELECTRONIC DICTIONARY
122
130
4
SECOND INPUT UNIT
DISPLAY
100
PROCESSOR
FIRST INPUT UNIT
SOUND COLLECTION UNIT
SOUND EMISSION UNIT
121
140
150
110
STORAGE UNIT
111
CONTENT
6B
ENGLISH-JAPANESE DICTIONARY B
ENGLISH-JAPANESE DICTIONARY A
DICTIONARY DATA
PRIORITY INFORMATION
THRESHOLD INFORMATION
6A
7
8
9

FIG. 3A

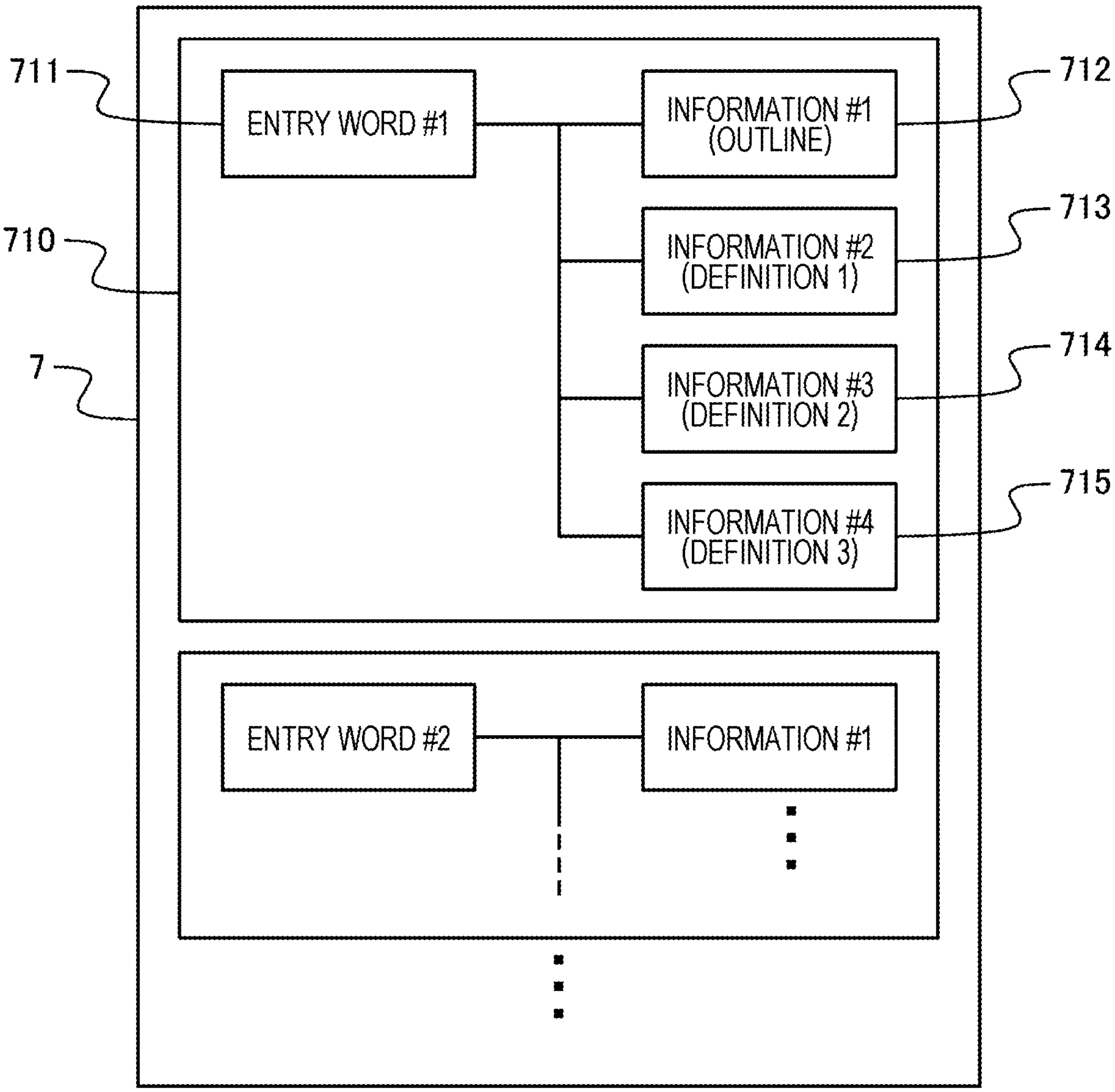

711
712
ENTRY WORD #1
INFORMATION #1 (OUTLINE)
713
710
INFORMATION #2 (DEFINITION 1)
7
714
INFORMATION #3 (DEFINITION 2)
715
INFORMATION #4 (DEFINITION 3)
ENTRY WORD #2
INFORMATION #1

FIG. 3B

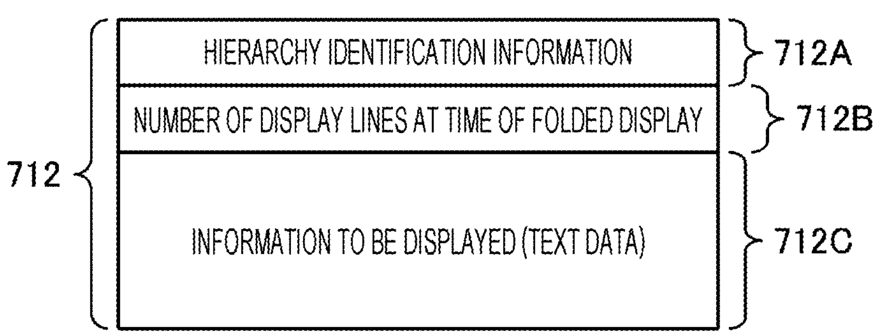

HIERARCHY IDENTIFICATION INFORMATION
712A
NUMBER OF DISPLAY LINES AT TIME OF FOLDED DISPLAY
712B
712
INFORMATION TO BE DISPLAYED (TEXT DATA)
712C

FIG. 4A

8A

PRIORITY		DISPLAY ORDER
HIGH	1	INFORMATION #1
	2	INFORMATION #2
	⋮	⋮
LOW	N	INFORMATION #N

FIG. 4B

8B

PRIORITY		DIFFERENCE OF HIERARCHY FROM INFORMATION CURRENTLY DISPLAYED
HIGH	1	1
	2	2
	⋮	⋮
LOW	N	N

FIG. 5A

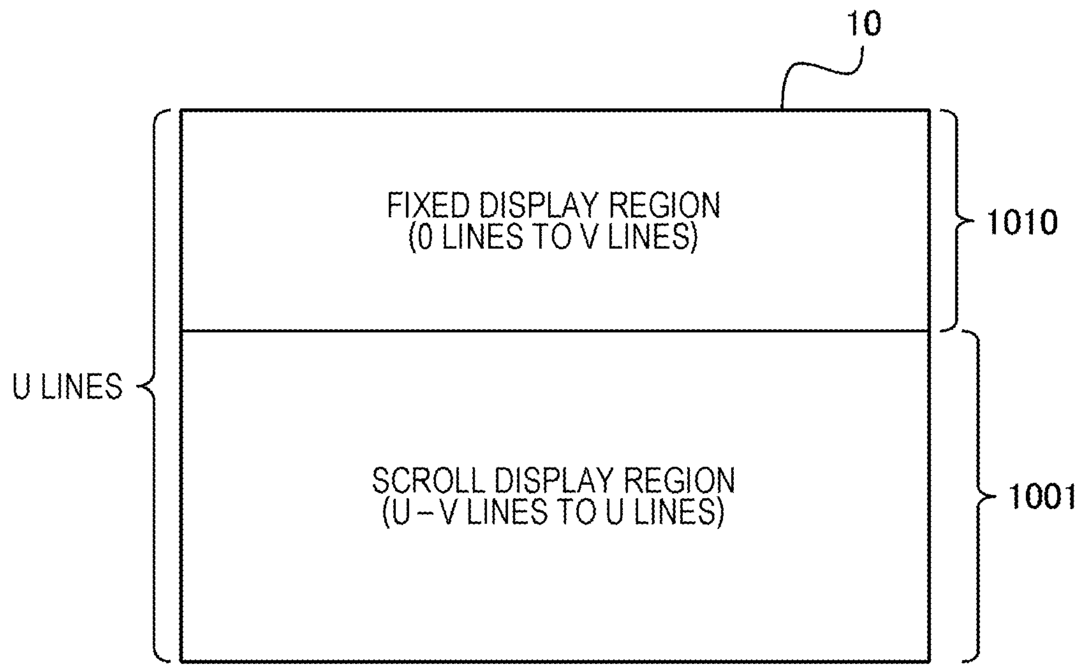

10
FIXED DISPLAY REGION
(0 LINES TO V LINES)
1010
U LINES
SCROLL DISPLAY REGION
(U−V LINES TO U LINES)
1001

FIG. 5B

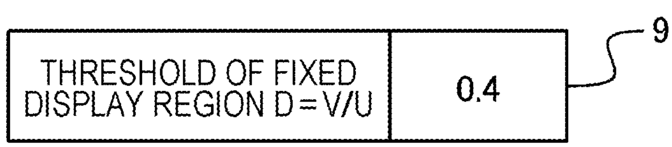

THRESHOLD OF FIXED
DISPLAY REGION D=V/U
0.4
9

FIG. 6

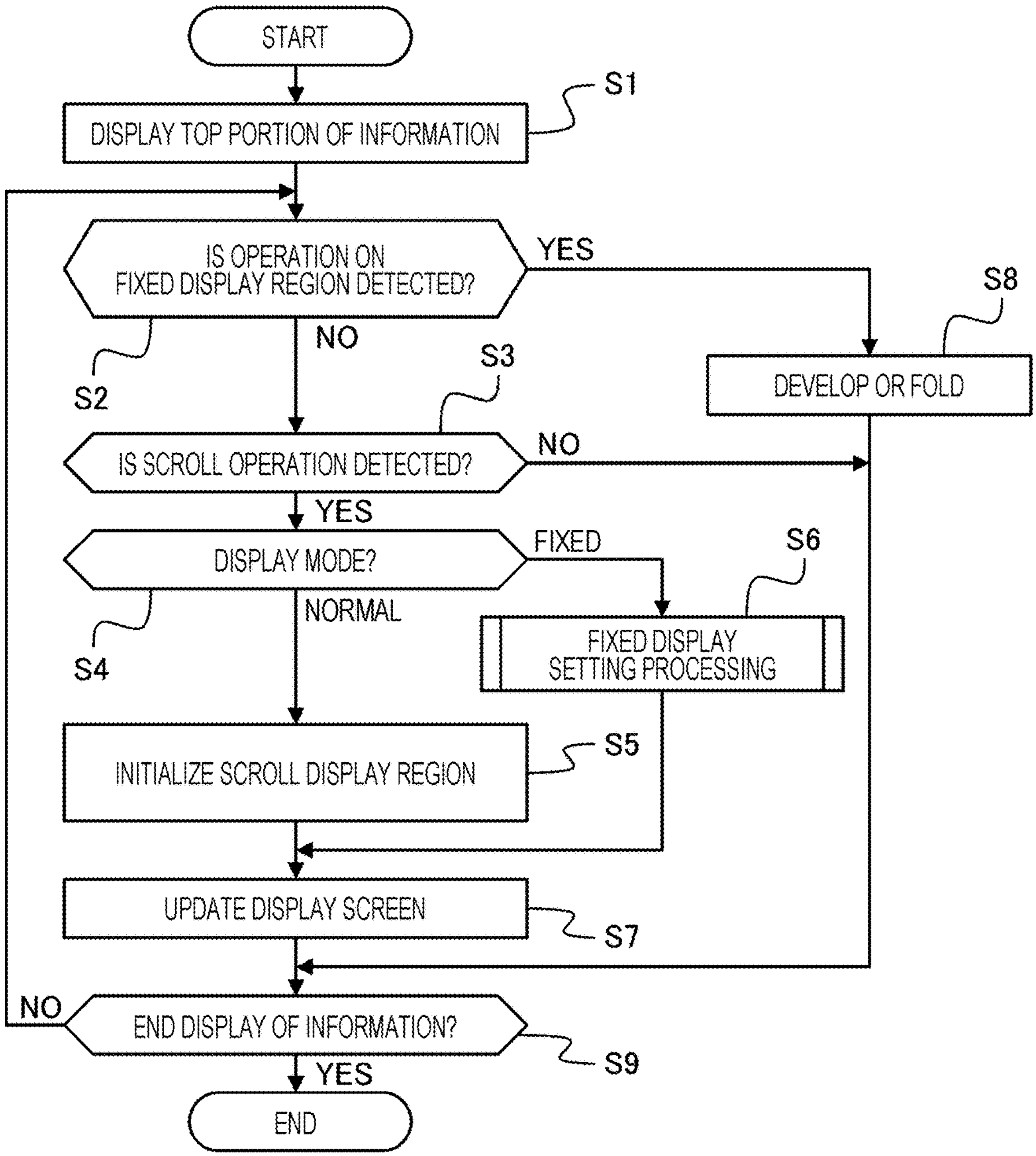

START
DISPLAY TOP PORTION OF INFORMATION
S1
IS OPERATION ON FIXED DISPLAY REGION DETECTED?
YES
NO
S2
S8
DEVELOP OR FOLD
S3
IS SCROLL OPERATION DETECTED?
NO
YES
DISPLAY MODE?
FIXED
NORMAL
S4
S6
FIXED DISPLAY SETTING PROCESSING
INITIALIZE SCROLL DISPLAY REGION
S5
UPDATE DISPLAY SCREEN
S7
NO
END DISPLAY OF INFORMATION?
S9
YES
END

FIG. 7

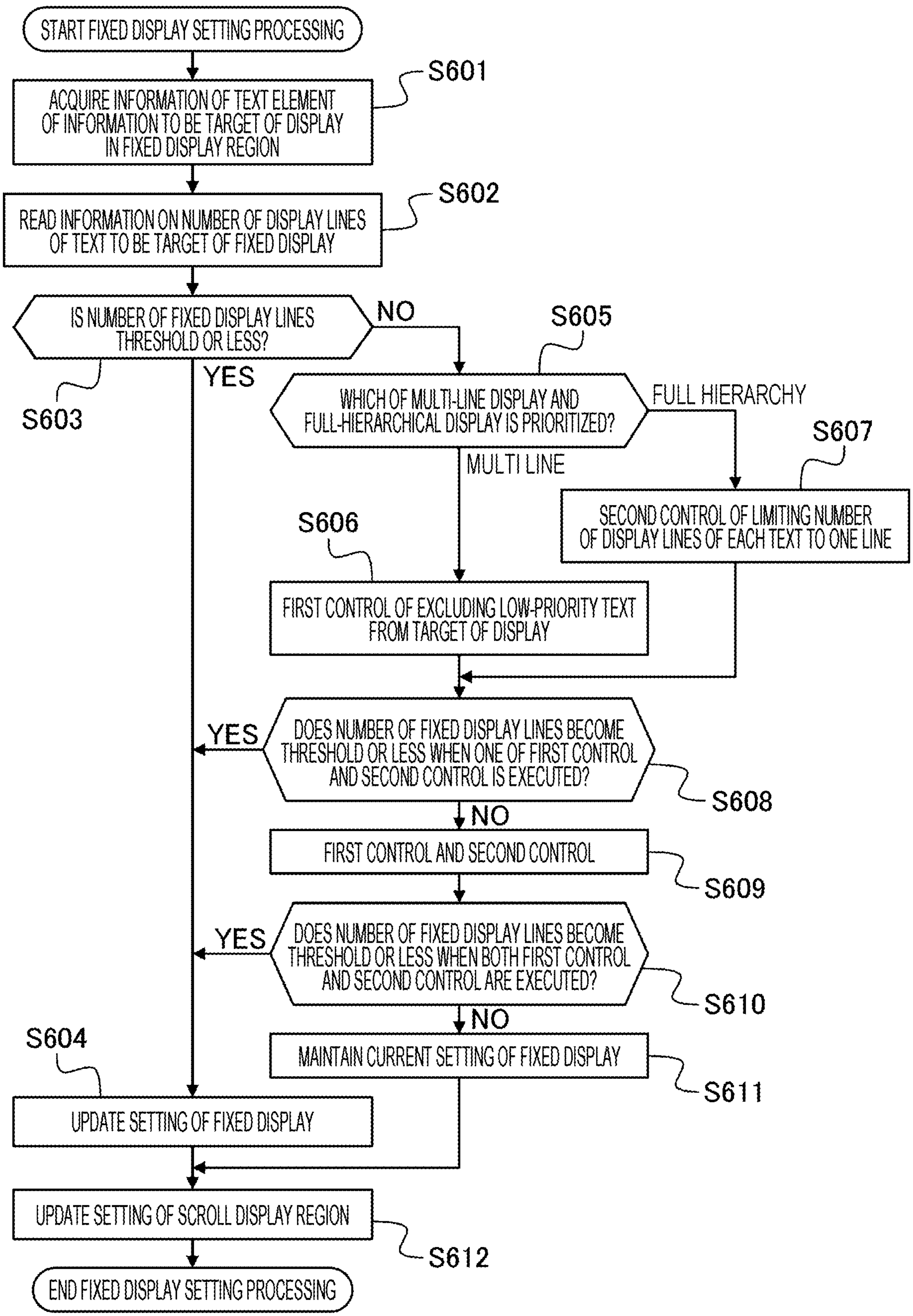

START FIXED DISPLAY SETTING PROCESSING
ACQUIRE INFORMATION OF TEXT ELEMENT OF INFORMATION TO BE TARGET OF DISPLAY IN FIXED DISPLAY REGION
S601
READ INFORMATION ON NUMBER OF DISPLAY LINES OF TEXT TO BE TARGET OF FIXED DISPLAY
S602
IS NUMBER OF FIXED DISPLAY LINES THRESHOLD OR LESS?
S603
NO
YES
S605
WHICH OF MULTI-LINE DISPLAY AND FULL-HIERARCHICAL DISPLAY IS PRIORITIZED?
FULL HIERARCHY
MULTI LINE
S607
SECOND CONTROL OF LIMITING NUMBER OF DISPLAY LINES OF EACH TEXT TO ONE LINE
S606
FIRST CONTROL OF EXCLUDING LOW-PRIORITY TEXT FROM TARGET OF DISPLAY
YES
DOES NUMBER OF FIXED DISPLAY LINES BECOME THRESHOLD OR LESS WHEN ONE OF FIRST CONTROL AND SECOND CONTROL IS EXECUTED?
S608
NO
FIRST CONTROL AND SECOND CONTROL
S609
YES
DOES NUMBER OF FIXED DISPLAY LINES BECOME THRESHOLD OR LESS WHEN BOTH FIRST CONTROL AND SECOND CONTROL ARE EXECUTED?
S610
NO
MAINTAIN CURRENT SETTING OF FIXED DISPLAY
S611
S604
UPDATE SETTING OF FIXED DISPLAY
UPDATE SETTING OF SCROLL DISPLAY REGION
S612
END FIXED DISPLAY SETTING PROCESSING

FIG. 8

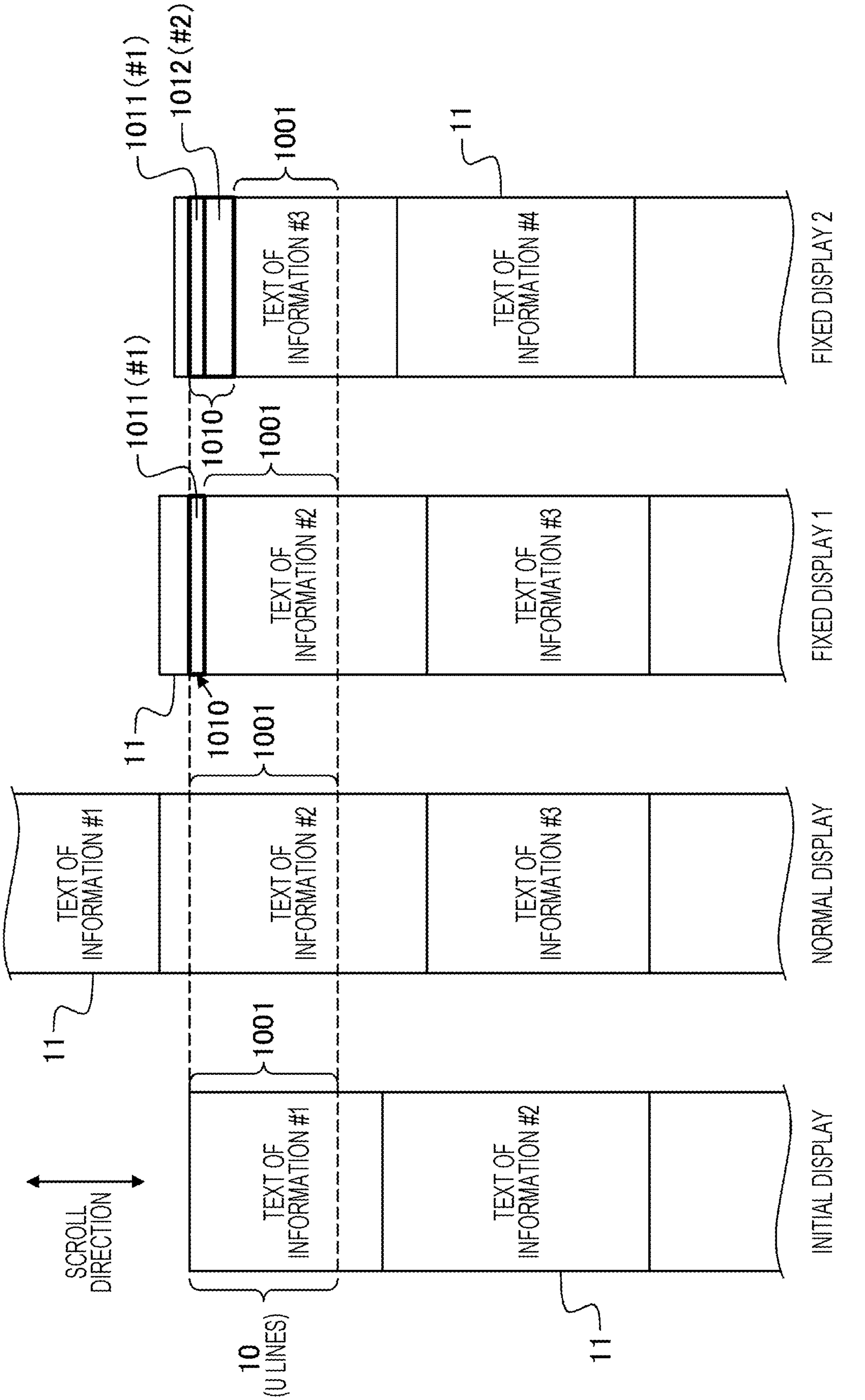

SCROLL DIRECTION
10 (U LINES)
11
1001
1010
1011 (#1)
1012 (#2)
TEXT OF INFORMATION #1
TEXT OF INFORMATION #2
TEXT OF INFORMATION #3
TEXT OF INFORMATION #4
INITIAL DISPLAY
NORMAL DISPLAY
FIXED DISPLAY 1
FIXED DISPLAY 2

FIG. 9A

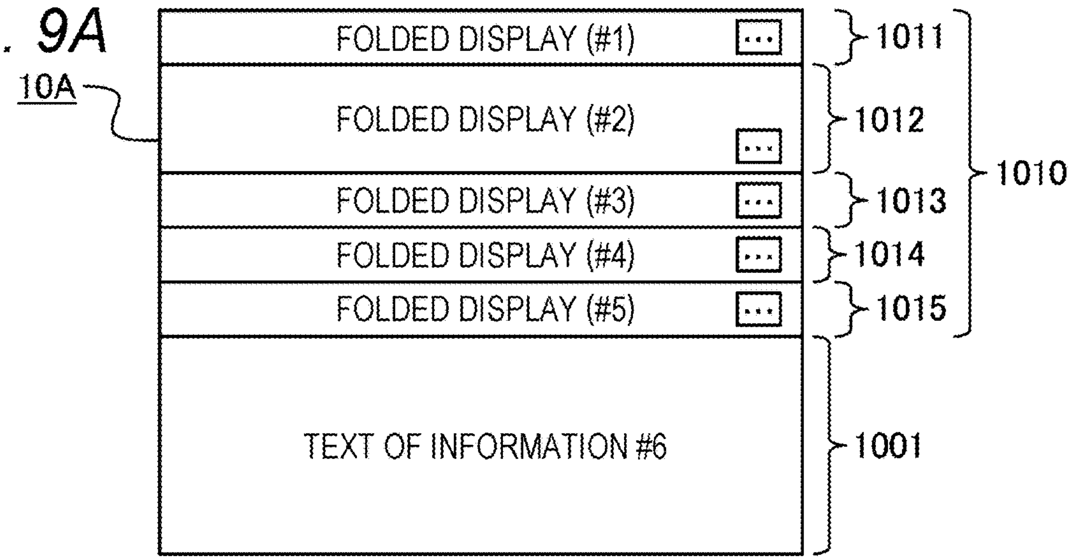


FIG. 9B

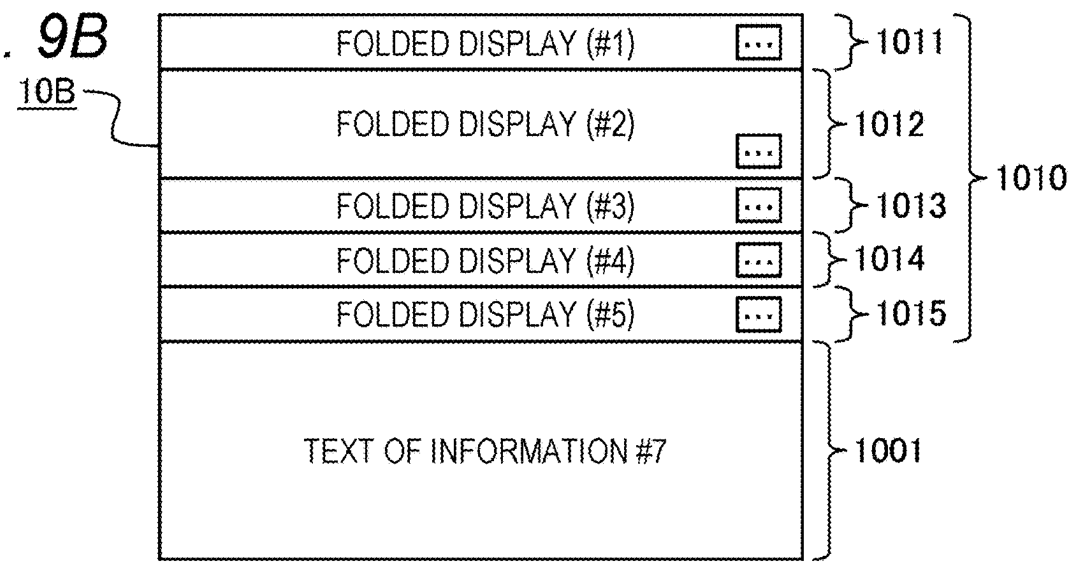


FIG. 9C

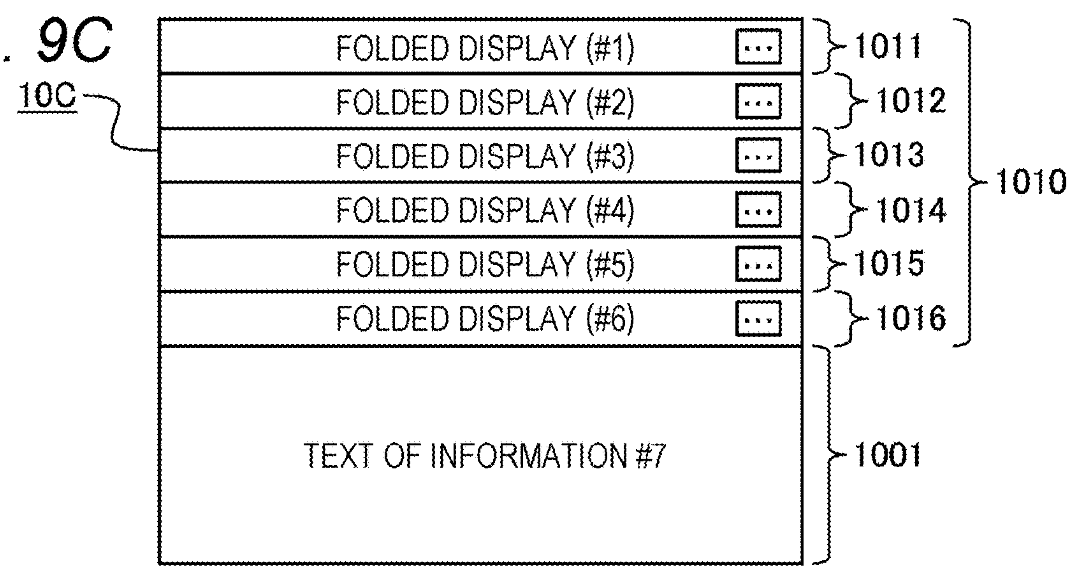


FIG. 10A

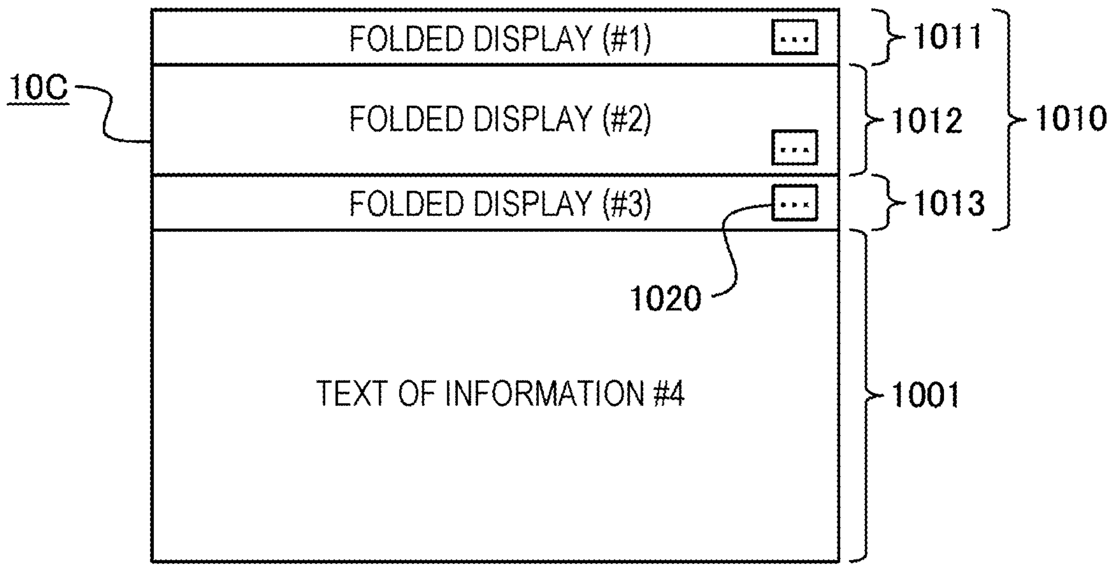
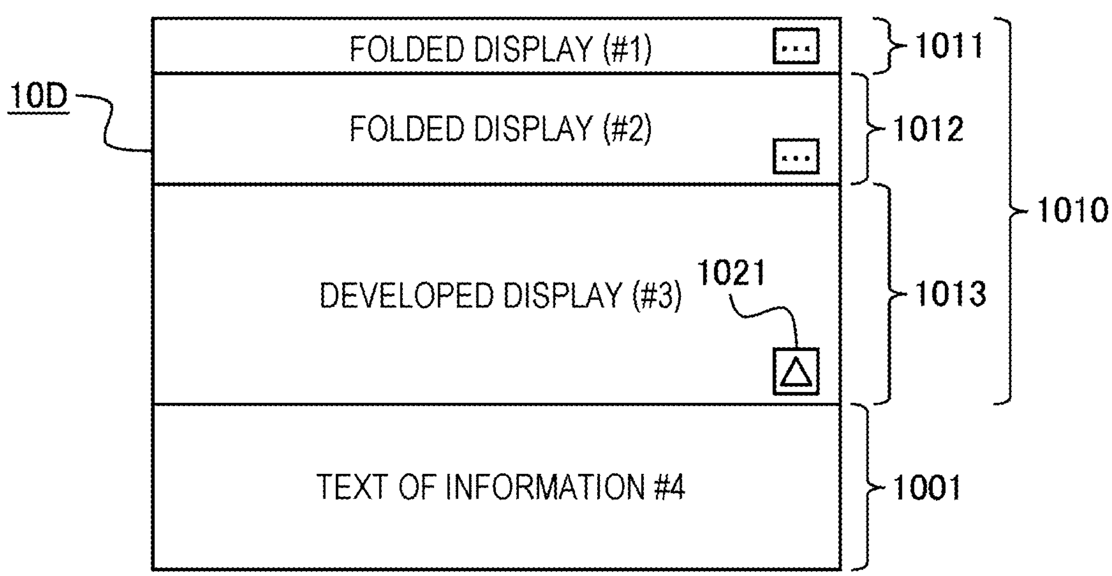


FIG. 10B



INFORMATION PROCESSING APPARATUS, DISPLAY CONTROL METHOD, AND RECORDING MEDIUM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority based on Japanese Patent Application No. 2024-030873 filed on Mar. 1, 2024, and the entire disclosure including the specification, claims, drawings, and abstract is incorporated herein by reference.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an information processing apparatus, a display control method, and a recording medium.

2. Related Art

[0003] In an electronic dictionary, in a case where all the detailed information related to a searched entry word cannot be displayed on a display at a time, a portion of the detailed information to be displayed on the display can be changed by a scroll operation. Some of this type of electronic dictionaries can divide a display screen of the display into upper and lower parts, and scroll one of the upper and lower divided screens in a state where the other of the divided screens kept in a fixed state (for example, JP 2013-186753 A).

SUMMARY

[0004] An information processing apparatus according to an aspect of the present disclosure includes a processor that, in a case of setting a first display region and a second display region on a display screen on which a display displays designated information, the first display region displaying a first portion of the information in a scrollable state and the second display region displaying a second portion of the information different from the first portion in a non-scrollable state, changes an amount of displayable information in the first display region and an amount of displayable information in the second display region in accordance with a user operation of scrolling display in the first display region.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a diagram illustrating an external configuration example of an electronic dictionary according to an embodiment;

[0006] FIG. **2** is a block diagram illustrating a functional configuration example of the electronic dictionary;

[0007] FIGS. **3**A and **3**B are diagrams for explaining an example of a hierarchical structure of dictionary data;

[0008] FIGS. **4**A and **4**B are diagrams for explaining an example of priority information;

[0009] FIGS. **5**A and **5**B are diagrams for explaining a threshold of a fixed display region in a display screen;

[0010] FIG. **6** is a flowchart illustrating an example of processing performed by the electronic dictionary;

[0011] FIG. **7** is a flowchart illustrating a specific example of fixed display setting processing in FIG. **6**;

[0012] FIG. **8** is a diagram illustrating an example of a change in a scrollable region in a display screen;

[0013] FIGS. **9**A to **9**C are diagrams for explaining an example of updating display in accordance with a scroll operation; and

[0014] FIGS. **10**A and **10**B illustrate an example of switching display of fixedly displayed information.

DETAILED DESCRIPTION

[0015] Hereinafter, an embodiment of the present disclosure will be described with reference to the drawings.

[0016] An electronic dictionary **1** illustrated in FIG. **1** includes a first housing portion **3** including a keyboard **2** and a second housing portion **5** including a touch panel display **4**. The first housing portion **3** and the second housing portion **5** are coupled to each other by, for example, a hinge mechanism or the like such that an angle of a display surface of the touch panel display **4** in the second housing portion **5** with respect to a surface of the first housing portion **3** on which keys of the keyboard **2** are disposed can be changed. Because the external configuration of the electronic dictionary **1** may be a well-known configuration, a specific description of the external configuration of the electronic dictionary **1** will be omitted in the present specification. Note that the external configuration of the electronic dictionary **1** is not limited to the external configuration illustrated in FIG. **1**.

[0017] The electronic dictionary **1** is an exemplification of an information processing apparatus according to the present disclosure. The electronic dictionary **1** can be an electronic device specialized in providing a plurality of types of content including dictionary content. As illustrated in FIG. **2**, from the viewpoint of the functional configuration, the electronic dictionary **1** includes a processor **100**, a storage unit **110**, a first input unit **121**, a second input unit **122**, a display **130**, a sound collection unit **140**, and a sound emission unit **150**. The electronic dictionary **1** may include a function not illustrated in FIG. **2**, for example, a communication unit for communicating with an external device.

[0018] The processor **100** controls the working of the electronic dictionary **1**. For example, the processor **100** executes the content of the dictionary, and performs processing of searching for an entry word input or selected by a user operation and displaying a search result. The functions of the processor **100** are provided by at least one processor that executes an operating system (OS) program, various programs related to the use of content stored (recorded) in the storage unit **110**, and the like. The processor can include a central processing unit (CPU).

[0019] The storage unit **110** stores content data **111** including a plurality of types of content, and other data, programs, and the like that are not illustrated. The storage device that provides the function of the storage unit **110** includes a read only memory (ROM) and a random access memory (RAM). A part of the functions of the storage unit **110** may be provided by, for example, a cache memory built in a processor such as a CPU, or a portable recording medium such as a card type memory.

[0020] The content data **111** can include content **6**A of an English-Japanese dictionary A and content **6**B of an English-Japanese dictionary B illustrated in FIG. **2**, and one or more pieces of content not illustrated in FIG. **2**. The contents **6**A and **6**B of the English-Japanese dictionary are examples of the content of the dictionary. The content of the dictionary may include content different from the English-Japanese dictionary, for example, a Japanese-English dictionary, a Japanese dictionary, a dictionary in which a relationship between a foreign language other than English and Japanese is described, and other character dictionaries and encyclopedias. The content of the dictionary includes, for example, dictionary data **7**, priority information **8**, and threshold information **9** illustrated in the content **6**A of the English-Japanese dictionary A in FIG. **2**.

[0021] The first input unit **121** and the second input unit **122** receive an input of information regarding an operation (working) of the electronic dictionary **1**. The input received by the first input unit **121** and the second input unit **122** includes an input for selecting the content and an input of the entry word or the like. The display **130** visualizes and displays information regarding the working and the like of the electronic dictionary **1**. The information displayed by the display **130** includes information on a content selection screen, information on an entry word input screen, information on a screen of the search result of the entry word, detailed information on the entry word, and the like. A function of the first input unit **121** is provided by, for example, the keyboard **2** (see FIG. **1**). Functions of the second input unit **122** and the display **130** are provided by the touch panel display **4** (see FIG. **1**). The second input unit **122** is an optional function (constitutional element) of the electronic dictionary **1** and may be omitted.

[0022] The sound collection unit **140** converts sound (sound wave) of the surroundings of the electronic dictionary **1** into voice data (electric signal). The sound emission unit **150** emits sound data (electric signal) as sound (sound wave) to the surroundings of the electronic dictionary **1**. In the electronic dictionary **1**, one or both of the sound collection unit **140** and the sound emission unit **150** may be omitted.

[0023] Next, the dictionary data **7**, the priority information **8**, and the threshold information **9** included in the content will be described.

[0024] The dictionary data **7** has, for every entry word, a hierarchical structure of an entry word and detailed information on the entry word. For example, in the dictionary data **7** illustrated in FIG. **3**A, first hierarchical structure information **710** for a first entry word (entry word #1) includes a root element **711** in the first hierarchy indicating the first entry word, and text elements **712** to **715** in the second hierarchy having a parent-child relationship with the root element **711**. In a case where the first entry word, which is the root element **711**, is an English word, the text element **712** includes, for example, text data of an outline (for example, information such as phonetic symbols, typical Japanese translations among Japanese translations, and word origins) for the first entry word. The text elements **713** to **715** includes, for example, text data in which a Japanese translation of the first entry word, an example sentence, and the like are described for every meaning of a word. The text elements **712** to **715** in the electronic dictionary **1** according to the present embodiment includes, for example, as in the text element **712** illustrated in FIG. **3**B, hierarchy identification information **712**A that identifies the text element **712**, a number of display lines **712**B at the time of folded display, and information (text data) **712**C to be displayed. The hierarchy identification information **712**A includes, for example, information indicating that the information is included in the second hierarchy under the root element **711** and information indicating the display order at the time when the text data of the text elements **712** to **715** is displayed. The number of display lines **712**B at the time of the folded display is the number of display lines of when text data of the text element **712** is abbreviated and displayed in a fixed display region (see FIG. **5**A) of the display screen. The number of text elements in the second hierarchy in one hierarchical structure information is not limited to four. Furthermore, the hierarchical structure information may include a text element in the third hierarchy under the text element in the second hierarchy. The element under the root element in the hierarchical structure information is not limited to the text element, and may include an element including text data and image data. The term “abbreviated display” in the present specification is intended to display a part of text data being a target of display in the fixed display region, or a keyword, a summary, or the like associated with the content of the text data in an allocated partial region in the fixed display region.

[0025] The description format of the information in the dictionary data **7** including the first hierarchical structure information **710** is not limited to a specific format. The information in the dictionary data **7** may be described, for example, in accordance with a description format of extensible markup language (XML).

[0026] The priority information **8** can be information indicating priority of when text data of a plurality of text elements is selectively displayed in a fixed display region of the display screen. In priority information **8**A illustrated in FIG. **4**A, the priority follows the display order of the text data to be displayed on the display **130**, and the text element including the text data with an earlier display order has a higher priority. The display order of the text data of the text elements **712** to **715** illustrated in FIG. **3**A is based on the information indicating the display order included in the hierarchy identification information of each text element. For example, the text data of the text element **712** is first displayed, and when an operation of scrolling in a predetermined direction (for example, a downward scroll operation of displaying a text below the displayed text in the text data) is performed, the text data of the text element **713**, the text data of the text element **714**, and the text data of the text element **715** are displayed in this order. In this case, according to the priority information **8**A, the text element **712** has the highest priority, and the text element **713**, the text element **714**, and the text element **715** have lower priority in this order. Note that the display order of the priority information **8**A in FIG. **4**A may also serve as information indicating the order of text data displayed in a scroll display region **1001** when an operation of scrolling in a predetermined direction is performed.

[0027] The priority information **8**B illustrated in FIG. **4**B follows the order of the difference in the hierarchy from the text element including the text data currently displayed (at the time of determining the priority), and the text element having a smaller difference has a higher priority. According to the priority information **8**B, at the time when the text data of the text element **714** among the text elements **712** to **715** illustrated in FIG. **3**A is displayed, the priority of the text element **713** is higher than that of the text element **712**.

[0028] The threshold information **9** can be an upper limit value of the fixed display region set in the display screen, and can be, for example, a value of V in the line **0** to the line V in FIG. **5**A. In the electronic dictionary **1** of the present embodiment, at the time when the text data of the plurality of text elements included in one piece of hierarchical structure information is displayed on the display **130**, as illustrated in FIG. **5**A, the display screen **10** can be vertically divided into two, the lower side can be a scroll display region **1001**, and the upper side can be a fixed display region **1010**. The scroll display region **1001** is a region in which a range of a text to be displayed on the display **130** can be changed in accordance with the scroll operation. The fixed display region **1010** is a region for displaying a text of a text element different from the text element including the text displayed in the scroll display region **1001** among the text (in other words, the text displayed in the scroll display region **1001** at the time when the user scrolls up) above the text displayed in the scroll display region **1001**. That is, the scroll display region **1001** is a scrollable region, and the fixed display region **1010** is a non-scrollable region.

[0029] In the electronic dictionary **1** of the present embodiment, a ratio between the scroll display region **1001** and the fixed display region **1010** in the display screen **10** changes in accordance with the user operation. As illustrated in FIG. **5**A, in a case where the display screen **10** can display U lines of text among all the text, the fixed display region **1010** is variable in a range of 0 lines to V (<U) lines, and the number of lines of the scroll display region **1001** is changed in accordance with the number of lines of the fixed display region **1010** in the display screen **10**. Note that the number of display lines of the text in the display screen **10** changes, for example, in accordance with the size of the characters. Therefore, the upper limit value (threshold) of the fixed display region in the threshold information **9** may be, for example, a value D obtained by dividing the maximum number of lines V of the fixed display region **1010** by the number of display lines U in the display screen **10** as illustrated in FIG. **5**B. In a case where U=10 and D=0.4, the fixed display region **1010** is variable in a range of 0 to 4 lines.

[0030] Note that, in the electronic dictionary **1** of the present embodiment, switching may be enabled between a working mode in which a ratio between the scroll display region **1001** and the fixed display region **1010** can be changed in accordance with the scroll operation and a working mode in which the entire display screen **10** is always set as the scroll display region **1001**.

[0031] Next, an example of display control of the text data in the electronic dictionary **1** of the present embodiment will be described with reference to FIGS. **6** and **7**. In a case where detailed information on a certain entry word is designated as information of the hierarchical structure to be displayed by a known user operation, the electronic dictionary **1** performs, for example, processing along flowcharts of FIGS. **6** and **7**. The processing described below is mainly performed by the processor **100** (for example, a processor that executes a program), and is performed in cooperation with the storage unit **110**, the input units **121** and **122**, and the display **130**.

[0032] First, the processor **100** causes the display **130** to display a top portion of text data (detailed information) that is information of the designated hierarchical structure (step S**1**). In a case where the text elements **712** to **715**, which are the detailed information on the entry word of the root element **711** illustrated in FIG. **3**A, are designated as the information on the hierarchical structure to be displayed, in step S**1**, the processor **100** causes the display **130** to display the display screen **10** including, for example, the top (first) text of the text data **712**C of the text element **712**. At this time, the entire display screen **10** is the scroll display region **1001**. In a case where the number of display lines of the text data **712**C is less than the number of displayable lines of the scroll display region **1001**, a portion of the text data of the text element **713** is displayed following the text data **712**C.

[0033] Next, the processor **100** determines whether or not an operation on the fixed display region **1010** has been detected by (step S**2**). The operation on the fixed display region **1010** can be, for example, the operation of pressing a button associated with the information displayed in the fixed display region **1010**.

[0034] If it is determined that the operation on the fixed display region **1010** has not been detected (step S**2**; NO), the processor **100** subsequently determines whether or not a scroll operation has been detected (step S**3**). If it is determined that the scroll operation has not been detected (step S**3**; NO), the processor **100** subsequently determines whether or not to end the display of the information (step S**9**). In step S**9**, the processor **100** determines to end the display of the information in the case of detecting an operation of hiding the entire information of the hierarchical structure currently displayed, for example, an operation of searching for another entry word, an operation of executing another content, or the like. If it is determined not to end the processing (step S**9**; NO), the processor **100** performs the determination in step S**2**.

[0035] If it is determined that the scroll operation has been detected (step S**3**; YES), the processor **100** determines whether the current display mode is “Normal” or “Fixed” (step S**4**). “Normal” is a display mode in which the fixed display region **1010** is not always displayed regardless of the range of the text displayed on the display screen **10**, and “Fixed” is a display mode in which the fixed display region **1010** corresponding to the range of the text displayed on the display screen **10** is set. If it is determined that the display mode is “Normal” (step S**4**; NORMAL), the processor **100** initializes the scroll display region **1001** of the display screen **10** (step S**5**). In step S**5**, the processor **100** sets the entire display screen **10** to the scroll display region **1001**. If it is determined that the display mode is “Fixed” (step S**4**: FIXED), the processor **100** performs fixed display setting processing (step S**6**). In step S**6**, the processor **100** sets the scroll display region **1001** and the fixed display region **1010** in the display screen **10**, and performs setting of the display in the fixed display region **1010**. After the processing of step S**5** or step S**6**, the processor **100** updates the display screen **10** on the basis of the processing result of step S**5** or step S**6** and the scroll amount corresponding to the detected scroll operation (step S**7**). After step S**7**, the determination in step S**9** described above is performed. If it is determined not to end the processing (step S**9**; NO), the processor **100** performs the determination in step S**2**.

[0036] On the other hand, if it is determined in step S**2** that the operation on the fixed display region **1010** has been detected (step S**2**; YES), the processor **100** unfolds or folds the information displayed in the fixed display region **1010** in accordance with the detected operation (step S**8**). A specific example of step S**8** will be described later. After step S**8**, the processor **100** performs the determination in step S**9** described above. If it is determined not to end the processing (step S**8**; NO), the processor **100** performs the determination in step S**2**. The processing in step S**8** may be incorporated into the fixed display setting processing in step S**6**.

[0037] The fixed display setting processing in step S**6** includes, for example, processing in steps S**601** to S**612** illustrated in FIG. **7**. A program for causing the electronic dictionary **1** to execute the fixed display setting processing may be incorporated as a part of a main routine or may be incorporated as a subroutine in a display control program corresponding to the flowchart in FIG. **6**.

[0038] In step S**601**, the processor **100** acquires information of the text element including information (text) to be the target of display in the fixed display region **1010** of the display screen **10** specified in accordance with the detected scroll operation. If the operation of scrolling down is detected, in step S**601**, the processor **100** acquires information of a text element including information (text) already displayed in the fixed display region **1010** and information of a text element including information (text) that can be newly displayed in the fixed display region **1010**. If the operation of scrolling up is detected, in step S**601**, the processor **100** identifies a text element including a text displayed in the uppermost line of the scroll display region **1001** at the time when the range of the text to be displayed in the scroll display region **1001** of the display screen **10** is changed in accordance with the detected scroll operation, and acquires information on a text element including a text displayed further on the top side than the text of the above-described text element.

[0039] Next, the processor **100** derives the number of display lines of information (text) to be the target of display in the fixed display region **1010** (step S**602**), and determines whether or not the derived number of display lines is equal to or less than a threshold (step S**603**). In step S**602**, the processor **100** calculates the sum of the number of display lines at the time of folded display of the text element specified as the target of display in the fixed display region **1010** in step S**601** among all the texts on the basis of the information of the text element acquired in step S**601**. In step S**603**, the processor **100** determines whether or not a value obtained by dividing the number of display lines derived in step S**602** by the number of display lines U of the display screen **10** is equal to or less than a threshold D of the threshold information **9** (see FIG. **5**B). If it is determined that the number of display lines is equal to or less than the threshold (step S**603**; YES), the processor **100** updates the setting of the fixed display (step S**604**). The setting of the fixed display includes setting of the range (the number of lines) of the fixed display region **1010** in the display screen **10**, the number of display lines of the text for every text element to be displayed in the fixed display region **1010**, and the range of the text to be displayed.

[0040] If it is determined that the number of display lines is larger than the threshold (step S**603**; NO), the processor **100** then determines which one of multi-line display and full-hierarchical display is prioritized (step S**605**). The "multi-line display" is intended to display the text of the text element in the number of display lines at the time of folded display, and the "full-hierarchical display" is intended to display the text of all the text elements specified in step S**601** as the target of display in the fixed display region **1010**. Which one of the multi-line display and the full-hierarchical display is prioritized may be switchable by a predetermined user operation. If it is determined that the multi-line display is prioritized (step S**605**; MULTI LINE), the processor **100** temporarily executes the first control of excluding the text element with low priority to be displayed in the fixed display region **1010** from the display target in the fixed display region **1010** on the basis of the priority information **8** (step S**606**). If it is determined that the full-hierarchical display is prioritized (step S**605**; FULL HIERARCHY), the processor **100** temporarily executes the second control to set the number of display lines of the text of all the text elements to be displayed in the fixed display region **1010** to one line (step S**607**). After step S**606** or S**607**, the processor **100** determines whether or not the number of display lines in the fixed display region **1010** becomes equal to or less than the threshold in a case where one of the first control and the second control is temporarily executed (step S**608**). The above "first" and "second" are intended only to distinguish between the two controls, and are not intended that the first control in step S**606** precedes (prioritizes) the second control in step S**607**. That is, the control in step S**607** may be referred to as the "first control", and the control in step S**606** may be referred to as the "second control".

[0041] If the number of display lines is equal to or smaller than the threshold as a result of temporarily executing the first control in step S**606** (step S**608**; YES), the processor **100** fully executes the first control of updating the setting of the fixed display on the basis of the text element to be displayed in the fixed display region **1010** excluding the text element excluded in step S**606**, and updates the setting of the fixed display (step S**604**). If the number of display lines is equal to or smaller than the threshold as a result of temporarily executing the second control in step S**607** (step S**608**; YES), the processor **100** fully executes the second control of setting the number of display lines of the text of all the text elements to be displayed in the fixed display region **1010** to one line, and updates the setting of the fixed display (step S**604**).

[0042] If it is determined that the number of display lines is larger than the threshold even after step S**606** or step S**607** is performed (step S**608**; NO), the processor **100** temporarily executes both the first control and the second control (step S**609**), and determines whether or not the number of display lines in the fixed display region **1010** becomes equal to or less than the threshold in a case where both the first control and the second control are temporarily executed (step S**610**). Step S**609** can be, for example, processing of limiting the number of display lines of the text of all the text elements to one line, the text elements remaining as the display target in the fixed display region **1010** by the processing of step S**606**. Step S**609** may be, for example, processing of further excluding a text element with low priority among all the text elements remaining as the display target by the processing in step S**606**, and thereafter, setting the number of display lines of the text of the text element remaining as the display target to one line.

[0043] If the number of display lines is equal to or smaller than the threshold as a result of performing step S**609** (step S**610**; YES), the processor **100** updates the setting of the fixed display such that the number of display lines of the text of all the text elements remaining as the display target in the fixed display region **1010** is set to one line (step S**604**). If it is determined that the number of display lines is larger than the threshold even after performing step S**609** (step S**610**; NO), the processor **100** maintains the current fixed display setting (step S**611**). The current fixed display setting is the setting of the fixed display at the time when the currently performed fixed display setting processing is started (in other words, the setting of the fixed display immediately before the detection of the scroll operation that triggers the start of the currently performed fixed display setting processing). After step S**604** or step S**611**, the processor **100**

updates the setting of the scroll display region **1001** on the basis of the setting of the fixed display (step **S612**), and ends the fixed display setting processing. Note that, if the information of the text element acquired in step **S601** is not changed before and after the detected scroll operation (in other words, in a case where the text displayed in the scroll display region **1001** before and after the scrolling is the text of the same text element), the processing in step **S602** and subsequent steps may be omitted.

[0044] A specific example of the display control processing described above with reference to FIGS. **6** and **7** will be described below. In step **S1**, for example, as in "INITIAL DISPLAY" in FIG. **8**, the entire display screen **10** having U lines is the scroll display region **1001**, and the text of the first (top) U lines of text data **11** in which the texts of the plurality of text elements are connected is displayed in the scroll display region **1001**. In the text data **11** in FIG. **8**, the upper side is the top side, and the lower side is the end side. Information #1, information #2, information #3, and information #4 in the text data **11** represent blocks corresponding to the text data of the text elements **712** to **715** in FIG. **3**A. In the text data **11** being initially displayed, a range to be displayed in the scroll display region **1001** can be changed by the scroll operation of scrolling down. In a case where the display mode of the electronic dictionary **1** is "Normal", even when the text displayed in the scroll display region **1001** is changed to the text of the information #2 by the scroll operation of scrolling down, the entire display screen **10** having U lines remains in the scroll display region **1001** as in "NORMAL DISPLAY" in FIG. **8**. The display screen **10** of the display **130** (touch panel display **4**) of the electronic dictionary **1** has a diagonal size of about several tens of centimeters, and there is a limit to information (text) that can be displayed in the display screen **10** at a time. In the case of the normal display, because the entire display screen **10** of the display **130** is set as the scroll display region **1001**, the browsability of the text of one text element is high, but it may be difficult to grasp the content of the text of the text element on the upper side (top side) of the text element of the currently displayed text.

[0045] In the electronic dictionary **1** according to the present embodiment, as described above, by fixedly displaying the display mode, the fixed display region **1010** can be set variable in the range of 0 lines to V (<U) lines in the display screen **10** having U lines. In the fixed display region **1010**, it is possible to perform the abbreviated display in which the text of the text element further on the top side of the text data **11** than the text element including the text displayed in the scroll display region **1001** is folded within a range of one line to several lines. Therefore, for example, when the text of the information #2 is displayed in the scroll display region **1001** as in "FIXED DISPLAY 1" in FIG. **8**, only the topmost line of the display screen **10** having U lines is set as the fixed display region **1010**, and the remaining U-1 lines become the scroll display region **1001**. At this time, a text **1011** (#**1**) obtained by folding the text of the information #1 into one line is displayed in the fixed display region **1010**. When the scroll operation is performed in such a display state, the display in the scroll display region **1001** is changed in accordance with the scroll operation, but the display of the folded text **1011** (**190** 1) in the fixed display region **1010** is not changed.

[0046] Further, when the information #3 is displayed in the scroll display region **1001** as in "FIXED DISPLAY 2" in FIG. **8**, for example, the text **1011** (#1) obtained by folding the text of the information #1 is displayed in the topmost line of the display screen **10** having U lines, and a text **1012** (#2) obtained by folding the text of the information #2 into two lines is displayed in two lines which are the second and third lines from the top. That is, in "FIXED DISPLAY 2" in FIG. **8**, three rows from the top of the display screen **10** having U lines become the fixed display region **1010**, and the remaining U-3 lines become the scroll display region **1001**. Note that, in "FIXED DISPLAY 2" in FIG. **8**, the folded text **1012** (#**2**) of the information #2 is set to have two lines, but the number of lines at the time of the abbreviated display may be one line or three lines. In a case where the upper side of the text data **11** is the top side and the lower side is the end side, the text that has been moved to the upper part of the display screen **10** by the downward scrolling and hidden is displayed by the abbreviated display in the fixed display region **1010** set in the upper part of the display screen **10**, so that the user can easily understand that the information displayed in the fixed display region **1010** is information above the information displayed in the scroll display region **1001**. Note that the position of the fixed display region **1010** on the display screen **10** is not limited to the upper part of the scroll display region **1001**. For example, in a case where the information displayed in the scroll display region **1001** is scrolled up to move to the lower part of the screen, the fixed display region **1010** may be set below the scroll display region **1001**. The text of each block in the text data **11** may be aligned in a display order assumed to be browsed while being scrolled up. For example, in text data **11** illustrated in FIG. **8**, the lower side may be the top side and the upper side may be the end side, and in this case, the fixed display region **1010** is set below the scroll display region **1001**.

[0047] Although the detailed description is omitted, in the display state of the fixed display **2** in FIG. **8**, by changing the display in the scroll display region **1001** from the text of the information #**3** to the text of the information #**2** by the upward scrolling, the display of the display **130** is changed from the display state of the fixed display **2** to the display state of the fixed display **1**. In the display state of the fixed display **1** in FIG. **8**, by changing the display in the scroll display region **1001** from the text of the information #**2** to the text of the information #**1** by the upward scrolling, the display of the display **130** is changed from the display state of the fixed display **1** to the display state of the initial display. Further, in the display state of the fixed display **1**, by performing the operation of changing the display mode from the fixed display to the normal display, the display of the display **130** is changed from the display state of the fixed display **1** to the display state of the normal display. On the other hand, in the display state of the normal display in FIG. **8**, by performing the operation of changing the display mode from the normal to the fixed, the display of the display **130** is changed from the display state of the normal display to the display state of the fixed display **1**.

[0048] As described above, in the electronic dictionary **1** according to the present embodiment, the ratio of the fixed display region **1010** in the entire display screen **10** is variable, and the text to be displayed in the fixed display region **1010** is folded in the range of one line to several lines. In addition, in the electronic dictionary **1** according to the present embodiment, an upper limit is set on the ratio of the fixed display region **1010** in the entire display screen **10** by the threshold information **9**. For example, even when the ratio of the fixed display region **1010** in a display screen **10**A illustrated in FIG. **9**A reaches the upper limit, the electronic dictionary **1** that has detected the scroll operation of scrolling down performs the fixed display setting processing illustrated in FIG. **7**. When the text to be displayed in the scroll display region **1001** is changed from the text of information #6 to the text of information #7 after the detected scroll operation, the number of text elements to be displayed in the fixed display region **1010** is increased by one. Therefore, when the text of all the target text elements is made to display in the fixed display region **1010**, the number of fixed display lines is increased by one or more lines as compared with that before the scroll operation. Therefore, the electronic dictionary **1** determines that the number of fixed display lines is larger than the threshold in step **S603**, and performs the temporary execution of the first control in step **S606** or the temporary execution of the second control in step S**607** through the determination in step S**605**.

[0049] In step S**606** performed in a case where the multi-line display is prioritized, the electronic dictionary **1** temporarily executes, for example, the first control of excluding the text having the lowest priority to be displayed in the fixed display region **1010** from the display target in the fixed display region **1010** from among the six texts, the information #1 to the information #6, on the basis of the priority information **8**. In a case where the priority information **8** is the priority information **8**A of the display order illustrated in FIG. **4**A, the text of the information #6 having the lowest priority is excluded from the display target in the fixed display region **1010**. As a result of step **S606**, the text to be displayed in the fixed display region **1010** remains five texts from the folded text **1011** of the information #1 to the folded text **1015** of the information #5 in the display screen **10**A in FIG. **9**A, and the number of display lines of each of the texts **1011** to **1015** folded and displayed does not change. Therefore, in step **S608** subsequent to step **S606**, the electronic dictionary **1** determines that the number of the fixed display lines becomes equal to or less than the threshold when the first control is executed, and fully executes the first control to update the fixed display setting (step **S604**). Note that, in this example, because the text to be displayed in the fixed display region **1010** does not change, the update of the setting of the fixed display in step **S604** is the update of overwriting in which the setting information does not change before and after step **S604**. That is, in a case where the display in the scroll display region **1001** is changed from the text of the information #6 to the text of the information #7 by the scroll operation, the screen displayed on the display **130** through the processing of step S**606** is changed from the display screen **10**A illustrated in FIG. **9**A to a display screen **10**B illustrated in FIG. **9**B.

[0050] In step S**607** performed in a case where the full-hierarchical display is prioritized, the electronic dictionary **1** temporarily executes the second control of setting the number of display lines of the six texts **1011** to **1016** of the information #1 to the information #6 to one line as in a display screen **10**C illustrated in FIG. **9**C. On the display screen **10**C, the folded text **1016** of the information #6 can be displayed by an amount obtained by reducing the number of display lines of the folded text **1012** of the information #2, the number of display lines of which is larger than the number of display lines of other folded texts, to one line on the display screen **10**A. Therefore, in step **S608** subsequent to step S**607**, the electronic dictionary **1** determines that the number of the fixed display lines becomes equal to or less than the threshold when the second control is executed, and fully executes the second control to update the fixed display setting (step **S604**). In this example, the setting of the fixed display is updated to the setting indicating that all the texts **1011** to **1016** of the information #1 to the information #6 are folded into one line and displayed.

[0051] Although the detailed description is omitted, when a scroll operation of scrolling up is performed while the display screen **10**B or the display screen **10**C is displayed, and the display of the scroll display region **1001** is changed from the text of the information #7 to the text of the information #6, the display of the display **130** is changed from the display screen **10**B or the display screen **10**C to the display screen **10**A. Further, for example, when the scroll operation of scrolling down is performed while the display screen **10**C is displayed and the display in the scroll display region **1001** is changed from the text of the information #7 to the text of the information #8, the electronic dictionary **1** maintains the current fixed display setting (step S**611**). In a case where the priority information **8** is the priority information **8**B illustrated in FIG. **4**B, when the text of the information #8 is displayed in the scroll display region **1001**, for example, the folded texts of the information #2 to #7 are displayed in the fixed display region **1010**.

[0052] As described above, when the user operation of scrolling the text (information) displayed in the scroll display region **1001** is detected, the electronic dictionary **1** according to the present embodiment changes the ratio between the scroll display region **1001** and the fixed display region **1010** in the display screen **10** in accordance with the user operation. In other words, the electronic dictionary **1** can change the amount of displayable information in the scroll display region **1001** and the amount of displayable information in the fixed display region **1010** in accordance with the scroll operation. Therefore, the electronic dictionary **1** according to the present embodiment can suppress a decrease in information browsability due to a decrease in the ratio of the scroll display region **1001** as compared with the electronic dictionary of JP 2013-186753 A in which the ratio of the fixed display region in the display screen is fixed. Note that, in the present specification, an example in which a text is folded (only information of first one line to several lines of the text is displayed) has been described as an example of the abbreviated display of the fixed display region **1010**, but the display form at the time of the abbreviated display is not limited thereto. The display form at the time of the abbreviated display in the fixed display region **1010** may be, for example, a form in which it is easy to intuitively understand what kind of information the text includes, such as displaying a keyword extracted from the text that is a target of the abbreviated display.

[0053] Further, in the electronic dictionary **1** of the present embodiment, the proportion of the scroll display region **1001** decreases as the amount of scroll from the display state corresponding to the initial display in FIG. **8** increases. However, by setting the upper limit (threshold) to the proportion of the fixed display region **1010**, more pieces of information can be displayed on the display screen **10** while decrease in the browsability of information displayed in the scroll display region **1001** is suppressed. Furthermore, by reducing the number of texts (text elements) to be displayed in the fixed display region **1010** or the number of display lines at the time of the folded display of each text on the basis of the priority information **8** such that the ratio of the fixed display region **1010** does not exceed the upper limit, deterioration of operability can be suppressed. In particular, by reducing the number of display lines at the time of the folded display of each text, texts of more text elements can be displayed in the fixed display region **1010**, and the decrease in the browsability of information is also suppressed. In addition, in a case where the number of texts (text elements) to be displayed in the fixed display region **1010** is reduced on the basis of the priority information **8**, because a text of a text element close to the text (text element) to be displayed in the scroll display region **1001** in accordance with the scroll operation remains in the fixed display region **1010**, the relationship between the text to be displayed in the scroll display region **1001** and the text to be displayed in the fixed display region **1010** becomes easily grasped.

[0054] Furthermore, as described above with reference to FIG. **6**, the electronic dictionary **1** according to the present embodiment may be able to unfold text data that is folded and displayed in the fixed display region **1010**. For example, FIG. **10**A illustrates a display screen **10**C in which the three texts **1011** to **1013** of the information #**1** to the information #3 are folded and displayed in the fixed display region **1010**. Operation buttons **1020** are displayed in association with the three texts **1011** to **1013** displayed in the fixed display region **1010**. When the operation button **1020** associated with the folded text **1013** of the information #3 is pressed while the display screen **10**C is being displayed, the electronic dictionary **1** determines that the operation on the fixed display region **1010** has been detected (step S**2**; YES). At this time, the electronic dictionary **1** changes the text **1013** of the information #3 from the folded display to the unfolded display (step S**3**) as in a display screen **10**D illustrated in FIG. **10**B, and displays an operation button **1021** for returning to the folded display. Upon detecting the operation of pressing the operation button **1021**, the electronic dictionary **1** causes the display **130** to display the display screen **10**C in which the text **1013** of the information #3 is changed from the unfolded display to the folded display.

[0055] As described above, by temporarily unfolding and displaying the text folded and displayed in the fixed display region **1010**, it becomes possible to save the trouble of performing the scroll operation on the scroll display region **1001**, and the operability of the electronic dictionary **1** is improved. Note that, in a case where the information to be displayed in the fixed display region **1010** includes a portion that cannot fit in the display screen **10**, the information displayed in the fixed display region **1010** in the unfolded state may be made temporarily scrollable.

[0056] The embodiments described above are specific examples to facilitate understanding of the present disclosure, and the present disclosure is not limited to the embodiments described above. The information processing apparatus and the display control method can be variously modified and changed without departing from the scope of the claims. For example, the information processing apparatus according to the present disclosure is not limited to a dedicated electronic device that provides the content of a dictionary such as the electronic dictionary **1** described above, and may be a smartphone, a tablet computer, or a computer in any form that can execute the content of another dictionary. For example, the information processing apparatus may be able to execute the content of the dictionary stored in a server device connected via a communication network such as the Internet by using a web browser. Furthermore, as a program that can cause a computer to execute a function similar to that of the above-described information processing apparatus, the program can be stored and distributed in a recording medium such as a memory card (such as ROM card and RAM card), a magnetic disk (such as flexible disk and hard disk), an optical disk (such as CD-ROM and DVD), or a semiconductor memory. Then, the computer reads the program recorded in the recording medium, and by the operation being controlled by the program, processing similar to the functions described in the embodiment can be realized.

What is claimed is:

1. An information processing apparatus comprising a processor, wherein

in a case of setting a first display region and a second display region on a display screen on which a display displays designated information, the first display region displaying a first portion of the information in a scrollable state and the second display region displaying a second portion of the information different from the first portion in a non-scrollable state,

the processor changes an amount of displayable information in the first display region and an amount of displayable information in the second display region in accordance with a user operation of scrolling display in the first display region.

2. The information processing apparatus according to claim **1**, wherein

the second portion of the information is a portion closer to a top side of the information than the first portion of the information, and

the processor moves the information displayed on the top side of the first portion to the second display region and displays the information in accordance with a user operation of scrolling the display of the first display region from the top side to an end side of the information.

3. The information processing apparatus according to claim **2**, wherein the processor displays the top side of the information on an upper side of the first display region, and in a state where the end side of the information is displayed on a lower side of the first display region and in a case where display of the first display region is scrolled from the top side toward the end side of the information, the processor sets the second display region as a region on the upper side of the first display region.

4. The information processing apparatus according to claim **2**, wherein the processor decreases the amount of displayable information in the first display region and increases the amount of displayable information in the second display region as a scroll amount from a top of the information to the first portion increases.

5. The information processing apparatus according to claim **2**, wherein, in a case where the amount of displayable information in the second display region is increased in accordance with a user operation, and in a case where the amount of displayable information after increase exceeds a threshold, the processor changes a display form of the information of the second portion to be displayed in the second display region such that the amount of the information of the second portion to be displayed in the second display region is equal to or less than the threshold.

6. The information processing apparatus according to claim **2**, wherein

the information is text data including a plurality of lines of text, and

the processor is configured to: fold text of a plurality of lines within a predetermined range displayed in the second display region among the text data such that a number of lines of the text becomes smaller than a number of lines of the plurality of lines; and

display a folded text subjected to the folding in the second display region.

7. The information processing apparatus according to claim **6**, wherein

the plurality of lines of the text of the text data is classified into a plurality of blocks, and

for each of the blocks displayed in the second display region, the processor is configured to:

fold, for every block, text of a plurality of lines such that the number of lines of the text becomes smaller than the number of lines of the plurality of lines; and

display a plurality of folded texts subjected to the folding in the second display region.

8. The information processing apparatus according to claim **7**, wherein the processor changes the number of the folded texts to be displayed in the second display region in accordance with the user operation of scrolling display in the first display region.

9. The information processing apparatus according to claim **7**, wherein the processor is configured to:

determine whether or not to prioritize display of all blocks to be displayed in the second display region in a case where the amount of displayable information in the second display region is increased in accordance with a user operation and in a case where the amount of displayable information after increase exceeds a threshold;

in a case where it is determined to prioritize the display of all the blocks, perform first control of reducing a display range of text at the time of folding such that the folded texts of all the blocks to be displayed in the second display region are displayed; and

in a case where it is determined not to prioritize the display of all the blocks, perform second control of excluding the folded text of a block having a priority order that is low among all the blocks to be displayed in the second display region from a display target in the second display region.

10. The information processing apparatus according to claim **9**, wherein the priority order is higher for a block closer to a block of the text displayed in the first display region.

11. The information processing apparatus according to claim **9**, wherein the processor performs both of the first control and the second control in a case where the amount of displayable information exceeds the threshold even when one of the first control and the second control is performed.

12. The information processing apparatus according to claim **10**, wherein in a case where the amount of displayable information exceeds the threshold even when both of the first control and the second control are performed, the processor maintains display in the second display region before the user operation is detected.

13. The information processing apparatus according to claim **6**, wherein the processor switches the display of the folded text between a state of a folded text and a state of an unfolded text in accordance with a user operation on the folded text.

14. The information processing apparatus according to claim **2**, wherein

the processor further performs control to switch between a first display mode in which display of the second display region is enabled and a second display mode in which the display of the second display region is disabled according to a user operation, and

in a case where the display mode is switched to the first display mode while the information is displayed in the second display mode, the processor determines the amount of displayable information in the first display region and the amount of displayable information in the second display region in accordance with a scroll amount from a top of the information to the first portion displayed in the first display region at a time point of the switching.

15. An information processing method executed by a computer of an information processing apparatus, the computer that performs,

in a case of setting a first display region and a second display region on a display screen on which a display displays designated information, the first display region displaying a first portion of the information in a scrollable state and the second display region displaying a second portion of the information different from the first portion in a non-scrollable state,

changing an amount of displayable information in the first display region and an amount of displayable information in the second display region in accordance with a user operation of scrolling display in the first display region.

16. A non-transitory storage medium storing a computer readable program of an information processing apparatus comprising a processor, the program causing the computer to execute processing of,

in a case of setting a first display region and a second display region on a display screen on which a display displays designated information, the first display region displaying a first portion of the information in a scrollable state and the second display region displaying a second portion of the information different from the first portion in a non-scrollable state,

changing an amount of displayable information in the first display region and an amount of displayable information in the second display region in accordance with a user operation of scrolling display in the first display region.

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